

KOINGNAAS, South Africa

WMO: 685130

Lat: 30.1956S

Lon: 17.2903E

Elev: 95

StdP: 100.19

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	6.5	-6.7	2.2	21.3	-4.1	2.7	21.0	11.9	15.7	11.1	17.4	4.0	60	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	31.5	16.8	29.2	16.1	27.1	15.7	20.1	25.4	19.4	24.1	18.8	23.1	6.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	21.4	17.8	12.9	20.8	17.2	12.4	20.5	58.1	25.6	55.5	24.2	53.7	23.2	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	11.3	10.2	DB	2.9	36.2	0.9	1.0	2.2	36.9	1.7	37.5	1.1	38.1	0.5	38.8	(4)
			WB	2.0	20.9	1.1	1.2	1.1	21.8	0.5	22.5	-0.2	23.1	-1.0	24.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	20.0	19.9	19.1	18.1	16.0	14.6	14.4	13.5	15.0	16.6	17.8	19.2	(6)
		DBStd	3.55	1.42	1.67	2.43	3.59	3.46	3.55	4.02	2.84	2.63	2.51	2.05	1.53	(7)
		HDD10.0	3	0	0	0	0	1	2	1	0	0	0	0	0	(8)
		HDD18.3	794	2	3	14	44	92	127	141	157	108	67	31	7	(9)
		CDD10.0	2560	310	278	283	244	187	138	137	109	149	204	235	284	(10)
		CDD18.3	309	53	47	39	39	22	14	18	7	8	13	16	33	(11)
		CDH23.3	1679	118	144	205	317	177	107	114	60	94	141	102	100	(12)
		CDH26.7	525	14	39	75	118	56	32	25	18	36	55	32	24	(13)
(14)	Wind	WSAvg	5.0	5.8	5.4	5.1	4.6	4.1	4.4	4.6	5.1	5.4	5.7	5.8		(14)
(15)	Precipitation	PrecAvg	123	5	3	5	7	12	17	19	20	10	7	9	3	(15)
(16)		PrecMax	168	16	11	20	24	40	52	62	81	44	33	67	9	(16)
(17)		PrecMin	67	0	0	0	0	0	1	2	0	0	0	0	0	(17)
(18)		PrecStd	28	5	3	6	7	10	15	15	18	10	8	16	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	28.4	31.7	32.7	33.5	31.3	30.2	29.4	29.7	31.2	32.8	30.5	30.8		(19)
(20)		MCWB	19.3	19.4	18.5	17.8	15.6	15.0	14.4	13.7	14.0	15.5	16.2	19.0		(20)
(21)		DB	25.6	26.8	28.9	30.1	27.7	26.6	26.6	24.6	26.1	27.8	26.0	25.4		(21)
(22)		MCWB	18.7	18.7	17.4	16.5	14.3	13.3	13.5	12.6	13.7	15.4	15.9	17.7		(22)
(23)		DB	24.4	24.6	25.2	27.0	24.6	23.3	23.8	20.7	22.1	23.6	23.2	23.9		(23)
(24)		MCWB	18.2	18.3	17.5	15.7	13.7	12.4	12.4	11.1	12.6	14.8	15.8	17.4		(24)
(25)		DB	23.4	23.5	23.1	24.1	21.7	20.4	20.8	17.9	19.2	20.8	21.6	22.7		(25)
(26)		MCWB	17.9	17.8	17.2	15.0	13.2	11.7	11.5	11.5	13.0	14.3	15.4	17.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	21.0	21.3	20.7	20.0	17.1	16.0	15.5	15.2	17.0	18.7	18.5	20.3		(27)
(28)		MCDB	25.0	27.0	26.7	29.5	22.2	24.0	24.0	22.1	23.6	27.3	24.1	25.4		(28)
(29)		WB	19.9	19.8	19.5	18.4	16.3	14.7	14.4	14.0	15.2	16.9	17.5	19.0		(29)
(30)		MCDB	23.7	24.3	25.0	26.3	21.4	19.6	21.1	18.9	21.1	22.3	22.2	23.5		(30)
(31)		WB	19.2	19.1	18.6	17.2	15.6	14.0	13.7	13.2	14.4	16.0	16.8	18.4		(31)
(32)		MCDB	22.7	23.3	23.3	23.1	20.2	18.3	19.3	17.3	19.2	20.4	21.2	22.4		(32)
(33)		WB	18.6	18.5	17.9	16.5	14.9	13.4	13.0	12.7	13.7	15.2	16.2	17.8		(33)
(34)		MCDB	22.1	22.4	22.0	21.3	18.9	17.2	17.8	16.1	17.8	19.2	20.2	21.5		(34)
(35)	Mean Daily Temperature Range	MDBR	7.8	8.0	8.4	9.3	9.7	9.8	10.2	9.1	8.9	8.8	8.4	7.9		(35)
(36)		5% DB	9.0	10.0	11.9	13.8	14.4	14.5	14.2	14.9	14.6	15.2	11.9	9.5		(36)
(37)		MCWBR	3.8	4.0	4.7	5.3	6.1	6.2	5.9	6.7	6.2	6.2	4.9	4.1		(37)
(38)		5% WB	7.7	8.3	8.9	10.2	10.1	10.1	11.1	10.0	10.2	9.9	9.3	7.9		(38)
(39)		MCWBR	3.9	4.0	4.4	5.1	5.6	5.9	6.2	5.7	5.3	5.1	4.6	3.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.345	0.334	0.317	0.301	0.292	0.300	0.317	0.333	0.343	0.328	0.332		(40)
(41)		taud	2.523	2.519	2.542	2.589	2.602	2.584	2.556	2.504	2.425	2.427	2.523	2.526		(41)
(42)		Ebn at Noon	1002	976	955	924	892	876	883	907	940	967	1005	1011		(42)
(43)		Edh at Noon	112	109	100	86	76	73	79	92	111	118	111	112		(43)
(44)	All-Sky Solar Radiation	RadAvg	8.32	7.60	6.42	4.91	3.75	3.24	3.55	4.35	5.65	7.04	8.09	8.61		(44)
(45)		RadStd	0.30	0.25	0.18	0.19	0.17	0.16	0.15	0.17	0.23	0.21	0.26	0.32		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page